Answer: D

**40** Background count rate =8.3 counts $s^{-1}$

Initial actual count rate =77.3 – 8.3 = 69 counts $s^{-1}$
After t=280 s, actual count rate = 61.6 – 8.3 = 53.3 counts $s^{-1}$

$$C = C_O e^{-\lambda t}$$

$$\ln C = \ln C_O - \lambda t$$

$$\lambda t = \ln\left(\frac{C_o}{C}\right)$$

$$\left(\frac{\ln 2}{T_{\frac{1}{2}}}\right)t = \ln\left(\frac{C_o}{C}\right)$$

$$T_{\frac{1}{2}} = \frac{(\ln 2)t}{\ln\left(\frac{C_O}{C}\right)} = \frac{\ln 2(280)}{\ln\left(\frac{69}{53.3}\right)} = 752s$$

Answer: B